

Credit Card Fraud Detection Using Machine Learning

Abstract

Credit card fraud has become one of the major challenges in the digital payment system. Fraudulent transactions can lead to huge financial losses for banks, businesses, and customers. The objective of this project is to build a Machine Learning model that can detect fraudulent credit card transactions effectively.

This project uses supervised machine learning algorithms such as Logistic Regression, Decision Tree, and Random Forest for fraud detection. The dataset used in this project contains anonymized transaction records. Since the dataset is highly imbalanced, proper preprocessing and evaluation techniques were applied.

The performance of the models was compared using accuracy score, confusion matrix, and classification report. Among all algorithms, the Random Forest Classifier achieved the highest accuracy.

Introduction

With the rapid growth of online transactions and digital banking systems, credit card fraud has increased significantly. Detecting fraudulent transactions manually is difficult because of the large volume of transactions happening every second.

Machine Learning provides intelligent techniques to identify suspicious patterns automatically. By training models on historical transaction data, fraud detection systems can predict whether a transaction is genuine or fraudulent.

The main goal of this project is to develop a machine learning-based system that can classify transactions as fraudulent or non-fraudulent.

Objectives

The objectives of this project are:

1. To analyze credit card transaction data.
2. To preprocess and clean the dataset.
3. To build machine learning models for fraud detection.
4. To compare the performance of different algorithms.
5. To identify the best-performing model.
6. To improve understanding of real-world AI and ML applications.

Problem Statement

Credit card fraud causes major financial losses every year. Traditional rule-based systems are not sufficient to detect modern fraudulent activities. Therefore, an intelligent automated fraud detection system is required.

This project focuses on building a machine learning model capable of detecting fraudulent credit card transactions accurately.

Dataset Description

The dataset used for this project was obtained from Kaggle.

Dataset Link:

<https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud>

Dataset Features

- The dataset contains transactions made by European cardholders.
- It includes both fraudulent and genuine transactions.
- Features are transformed using PCA for privacy reasons.
- The target variable is:
 - Class = 0 → Normal Transaction
 - Class = 1 → Fraudulent Transaction

Important Columns

- Time
- V1 to V28
- Amount
- Class

Technologies Used

Programming Language

- Python

Libraries Used

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Pickle

Development Environment

- Jupyter Notebook

Methodology

The following steps were performed during the project:

1. Data Collection

The dataset was downloaded from Kaggle and loaded into Jupyter Notebook.

2. Data Preprocessing

- Checked dataset shape
- Checked missing values
- Performed feature scaling using StandardScaler

3. Exploratory Data Analysis

- Analyzed class distribution
- Visualized fraud and normal transactions using graphs

4. Train-Test Split

The dataset was divided into:

- 80% Training Data
- 20% Testing Data

5. Model Training

Three machine learning algorithms were trained:

- Logistic Regression
- Decision Tree
- Random Forest

6. Model Evaluation

Models were evaluated using:

- Accuracy Score
- Confusion Matrix
- Classification Report

7. Model Comparison

The performance of all algorithms was compared to identify the best model.

Algorithms Used

Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for binary classification problems. It predicts whether a transaction is fraudulent or genuine.

Decision Tree Classifier

Decision Tree creates a tree-like structure using conditions and rules. It helps classify transactions based on feature values.

Random Forest Classifier

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting.

Results and Discussion

The following accuracy scores were obtained:

Model	Accuracy
Logistic Regression	0.999157
Decision Tree	0.999140
Random Forest	0.999596

Observation

- All models performed well on the dataset.
- Random Forest achieved the highest accuracy.
- Random Forest provided better fraud detection performance compared to the other algorithms.

Advantages of the Project

1. Detects fraudulent transactions automatically.
2. Reduces financial losses.
3. Improves transaction security.
4. Provides fast and accurate predictions.
5. Demonstrates real-world AI application.

Limitations

1. Dataset is highly imbalanced.
2. Real-world fraud patterns may change over time.
3. Model performance depends on data quality.
4. Large datasets require more computational power.

Future Enhancements

1. Use Deep Learning techniques.
2. Implement real-time fraud detection.
3. Deploy using Streamlit or Flask.

4. Improve performance using SMOTE.
5. Integrate with banking applications.

Conclusion

This project successfully implemented machine learning techniques for credit card fraud detection. Different classification algorithms were trained and evaluated using the Kaggle dataset.

Among all models, the Random Forest Classifier achieved the best performance with the highest accuracy. The project demonstrates how machine learning can help financial institutions detect fraudulent transactions effectively.

This project improved practical knowledge in data preprocessing, machine learning model building, evaluation, and real-world AI applications.

References

1. Kaggle Credit Card Fraud Detection Dataset
<https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud>
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